

$\text{map}(f, v) \rightarrow \text{list}$ $\gg \text{map}(\text{lambda } x: x+1, [2,3])$
 $[3,4]$

$\text{filter}(f, v) \rightarrow \text{list}$ $\gg \text{filter}(\text{lambda } x: x\%2==0, [2,3])$
 $[2]$

$\text{reduce}(f, v, s) \rightarrow \text{int}$ $\gg \text{reduce}(\text{lambda } x, y: x+y, [2,3], 0)$
 $0+1 \rightarrow 1$
 $x \quad y \quad 1+2 \rightarrow 3$
 $\quad \quad \quad 3+3 \rightarrow 6$

$\text{iterate}(f, x, n) \rightarrow \text{int}$ $\gg \text{iterate}(\text{lambda } x: x+1, 2, 3)$
 5

$\text{fixpoint}(f, x) \rightarrow \text{int}$ $\gg \text{fixpoint}(\text{lambda } x: x-x-2, 1)$
 $1 \rightarrow -1 \rightarrow \dots$

$\text{def swap_two}(v: \text{list}[\text{int}]) \rightarrow \text{list}[\text{int}]$
 $\gg$ sorts the first two unsorted elements from a list

if $\text{len}(v) < 2$:
 return v

elif $v[0] > v[1]$:
 return v

else:
 return $[v[0] + \text{swap_point}(v[1:2])]$

$\text{def bubblesort}(v: \text{list}[\text{int}]) \rightarrow \text{list}[\text{int}]$
 return $\text{fixpoint}(\text{swap_two}, v)$

$\text{def gcd}(tuple[\text{int}, \text{int}]) \rightarrow \text{tuple}[\text{int}, \text{int}]$

$m, n = t$

if $m == n$:
 return t

elif $m > n$:
 return $(m-n, n)$

else:
 return $(m, n-m)$

$\text{def gcd}(m: [\text{int}], n: [\text{int}]) \rightarrow \text{int}$:
 return $\text{fixpoint}(\text{gcd}, [0])$

$\text{def factorial}(n: \text{int}) \rightarrow \text{int}$:
 $\gg$ the factorial of n

return $\text{reduce}(\text{lambda } x, y: x \cdot y, \text{range}(1, n+1))$

$\text{def count}(n: \text{int}, v: \text{list}[\text{int}]) \rightarrow \text{int}$:
 $\gg$ counts the occurrences of n in v

return $\text{reduce}(\text{lambda } x, y: x+1, \text{filter}(\text{lambda } n: n==v, v), 0)$

list comprehension.

- $\text{Range}(m, n, k)$
- $\text{map}(f, v) [fx \text{ for } x \text{ in } \text{Range}(n)]$
- $\text{filter}(f, v) [fx \text{ for } x \text{ in } \text{Range}(n) \text{ if } n\%2==0]$

$[1, 2, 3]$
 $+ [4, 5, 6]$ $\text{map}(\text{lambda } xy: x+y, [1, 2, 3], [4, 5, 6])$

$[5, 7, 9]$

$\text{map}(\text{lambda } v, y: v+[y], [1], [2], [3])$

$[4, 5, 6]$

$[1, 4], [2, 5], [3, 6]$